European Commission

Innovation in the EU: technological and creative performances in numbers across urban and rural regions

2026

HIGHLIGHTS

- Innovation metrics developed, for the first time, at a sufficiently granular territorial level to capture differences between EU urban and rural regions provide a nuanced picture of Europe's technological and creative landscape.
- Research & Development (R&D) investment across EU regions remains highly concentrated, with urban regions investing around 2.4% of their GDP in R&D, compared with 1.6% in rural regions. A group of rural regions nevertheless exceeds 3% of GDP invested in R&D, reflecting strong local specialisation or the presence of public research institutions supporting innovation capacity.
- Patent activity is geographically concentrated, with urban regions recording about 510 applications per 100,000 inhabitants, and rural ones about 180. Yet, several rural regions, especially in Germany, Austria, and Finland surpass the EU average, supported by specialised industrial clusters and strong urban–rural linkages.
- Trademarks – reflecting branding efforts often linked to new product launches and market expansion – are also highly clustered. Small, open economies such as Luxembourg, Cyprus, Estonia or Austria lead on a per capita basis, with stocks of EU trademarks ranging from 1,400 to 440 per 100,000 inhabitants. Urban regions hold on average 346 valid trademarks per 100,000 inhabitants, compared with around 101 in rural regions, with rural outliers like Åland (FI), Traunviertel (AT), and Teruel (ES) showing the strength of niche industries and cross-border business ecosystems.
- Industrial designs, which highlight the creative and product-development dimension of innovation, show a strong territorial gradient. Urban regions hold on average 156 EU designs per 100,000 inhabitants, compared with 56 in rural regions. Yet, several rural territories, including Steyr-Kirchdorf (AT), Pirmasens and Karlsruhe-Stadt (DE), Vestjylland (DK), Tarnowski (PL) and Vercelli (IT) perform well above the EU average, reflecting strong craft-based sectors and specialised manufacturing traditions.

Introduction

Innovation plays a central role in advancing the European Union's key objectives for competitiveness and sustainability, while potentially helping regions adapt to structural and demographic change (EC, 2020; 2022). Although R&D expenditure has increased over recent years, reaching 2.2% of GDP in 2022, employing 4.6 million R&D personnel (Eurostat, 2024), the EU still falls short of the long-standing objective of investing 3% of GDP in R&D (European Commission, 2010). At the same time, persistent territorial disparities remain a major challenge, with several rural and peripheral areas facing structural barriers that limit their capacity to benefit from innovation-driven growth, reinforcing regional divides in competitiveness, job creation and opportunities.

The EU has developed a coordinated policy agenda to support more inclusive and place-based innovation ecosystems. Cohesion Policy 2021–2027 promotes territorial equity through smart specialisation strategies, R&D infrastructure and innovation networks. The *New European Innovation Agenda* (European Commission, 2022) seeks to accelerate and strengthen innovation across the EU and address the innovation divide by boosting R&D investment, technology transfer and local capacities. At the same time, the *European Innovation Ecosystems (EIE)* strand of *Horizon Europe* (European Commission, 2025) provides targeted support for regions with lower innovation performance, rural areas and less-developed territories, while also fostering deep-tech scale-ups, experimentation spaces, public procurement for innovation and the integration of SMEs into global value chains. In parallel, the *Common Agricultural Policy* (2023–2027) provides dedicated support to innovation in agriculture and the bioeconomy.

Despite the recognised potential of innovation to reduce territorial socio-economic divides, the evidence base has remained limited. Until recently, data on innovation performance were rarely available at a sufficiently granular level to distinguish between rural, intermediate and urban regions, nor were indicators consistently comparable across Member States. As a result, the scale and geography of the innovation gap could not be fully assessed, and many rural strengths and potential opportunities remained largely invisible to policymakers. Developing suitable, territorially-sensitive metrics is therefore essential to support more effective, place-based interventions and to ensure that all regions can contribute to and benefit from innovation-driven growth.

Within this framework, the *Startup Village Forum (SVF)* initiative, coordinated by the European Commission's Joint Research Centre, has been developing new metrics and studies on innovation and entrepreneurship in rural areas, while engaging with policymakers, practitioners, and local stakeholders to share knowledge, exchange experiences, and identify concrete actions to build and strengthen rural entrepreneurial and innovation ecosystems. The SVF is embedded in two core European Commission's strategic frameworks. Within the *Long-Term Vision for the EU's Rural Areas* (European Commission, 2021), it supports theflagship action on Research and Innovation for Rural Communities, highlighting the role of innovation in unlocking development opportunities in rural areas. At the same time, under the *New European Innovation Agenda* (European Commission, 2022) it contributes to the flagship action on Addressing the Innovation Divide, helping to extend the benefits of innovation-driven growth to all regions.

Innovation performances across EU regions by urban–rural typology

	URBAN REGIONS	INTERMEDIATE REGIONS	RURAL REGIONS	EU
R&D expenditure as % of GDP	**2.4%**	**2%**	**1.6%**	**2.2%**
Patent applications per 100,000 inhabitants	**510**	**350**	**180**	**380**
EU trademarks per 100,000 inhabitants	**346**	**182**	**101**	**235**
EU industrial designs per 100,000 inhabitants	**156**	**118**	**56**	**123**

Note: R&D figures are JRC estimates based on 2022 data from Eurostat and ARDECO. Patent figures reflect applications by EU-based inventors recorded in the 2023 Autumn edition of the PATSTAT database and cover filings since 1990. Trademark and design figures refer to stocks of valid EU Trade Marks and Registered EU Designs held by EU-based applicants, as of June 2025.
Source: JRC calculations and analysis based on Eurostat, ARDECO, EPO PATSTAT and EUIPO data.

This brief presents selected findings from the JRC's ongoing work, introducing for the first time a set of indicators constructed at a sufficiently granular territorial level to capture differences between urban and rural regions. It focuses on three complementary aspects: R&D expenditure as a key measure of investment in knowledge creation; patent applications capturing technological innovation efforts; and trademark and design applications reflecting branding, product differentiation and creative activities. Using data from Eurostat (2022), the European Patent Office (EPO-PATSTAT, 2023), and European Union Intellectual Property Office (EUIPO, 2025), the analysis maps territorial patterns at NUTS3 level, distinguishing between rural, intermediate and urban regions. These findings aim to support data-driven, place-based innovation policies that foster competitiveness, sustainability and a more balanced development across EU regions.

Where does R&D investment happen?

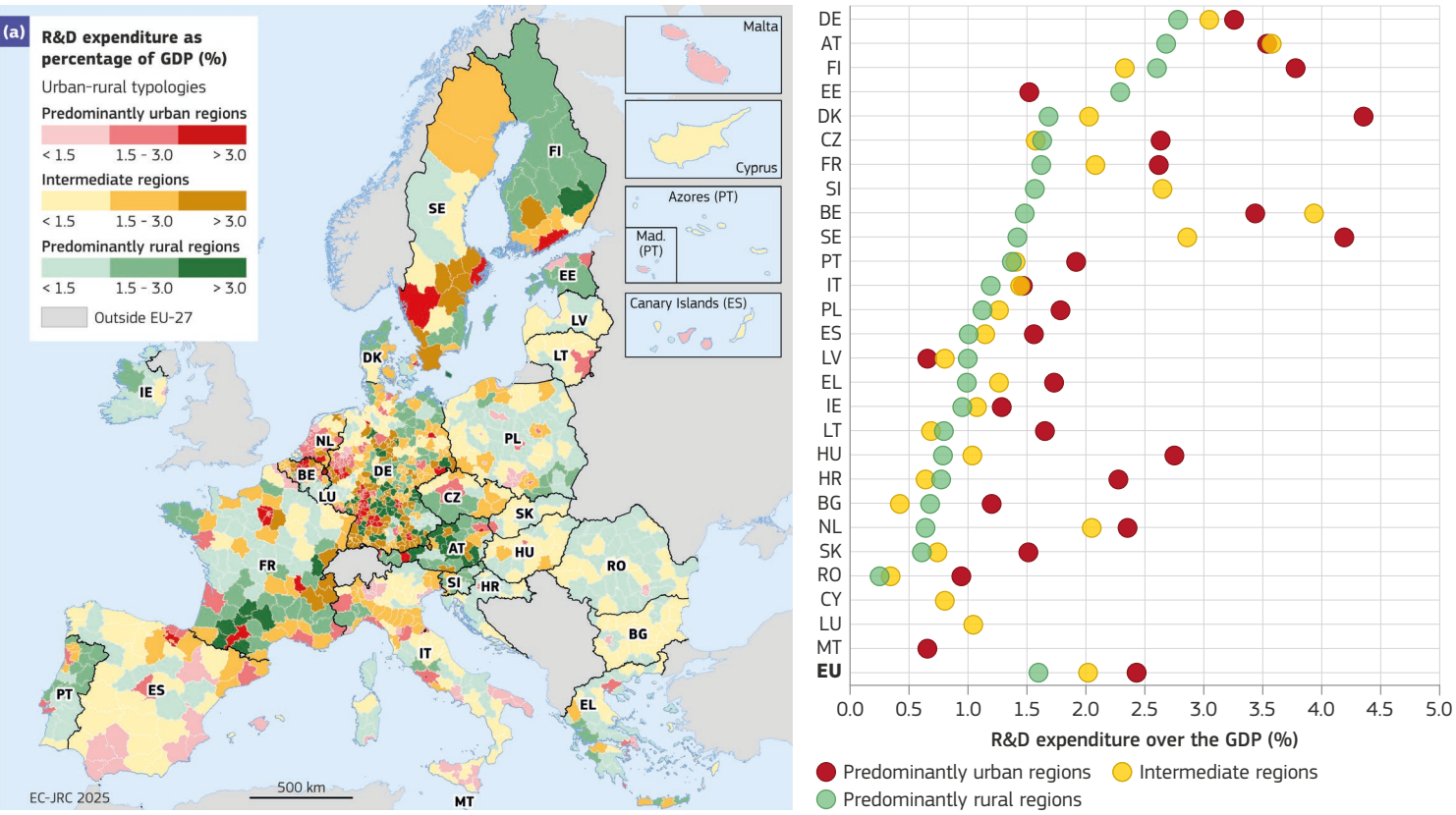
R&D expenditure provides a core measure of the efforts invested in knowledge creation and technological innovation capacity. In 2022, EU-27 regions invested an estimated EUR 313.6 billion PPS in R&D, with the business sector contributing nearly twice as much as the public sector (EUR 207.6 billion versus EUR 106.6 billion PPS). This corresponds to an average R&D intensity of around 2.2% of EU GDP, although substantial differences remain across Member States. Leading performers such as Belgium, Sweden, Austria, Germany, Finland and Denmark exceed 3% of GDP, driven primarily by strong business-sector participation. Conversely, R&D intensity remains below 1% of GDP in Romania, Bulgaria, Latvia, Cyprus and Malta.

Urban regions lead R&D investment, with notable rural outperformers

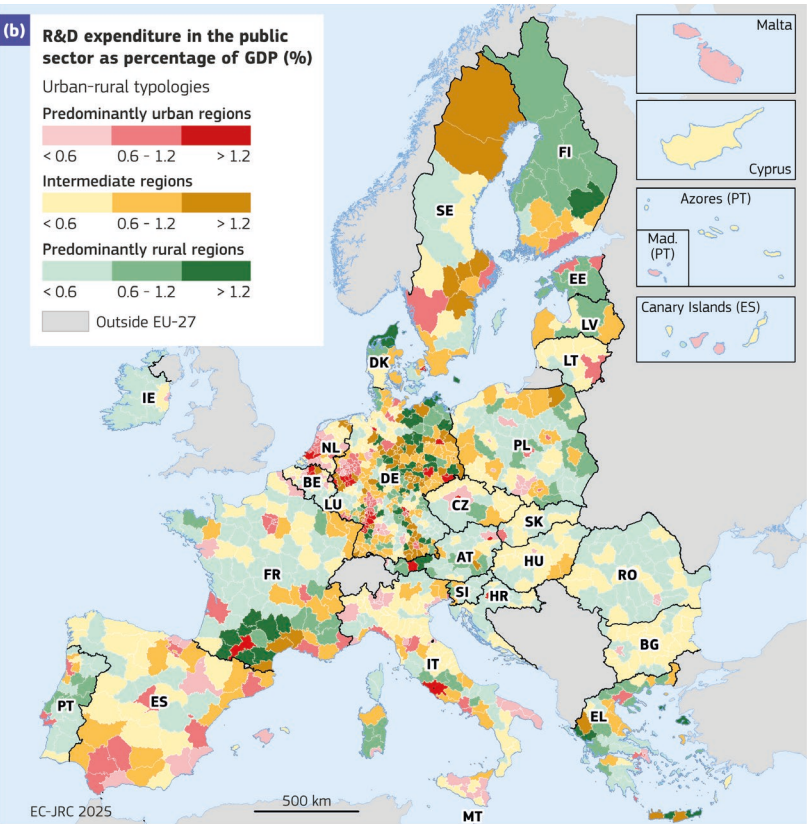
The geography of R&D investment shows a pronounced rural–urban divide. Accounting for 56% of total R&D expenditure, **urban regions invest approximately 2.4% of their GDP in R&D, while rural regions invest on average 1.6%.**[1] This underscores the key role of large urban centres (e.g., Stuttgart, Brussels, Innsbruck, Stockholm, Toulouse, Berlin, Vienna, Helsinki) as hubs where research capacity and technological development activities are concentrated, and highlights persistent gaps in innovation inputs across territorial types.

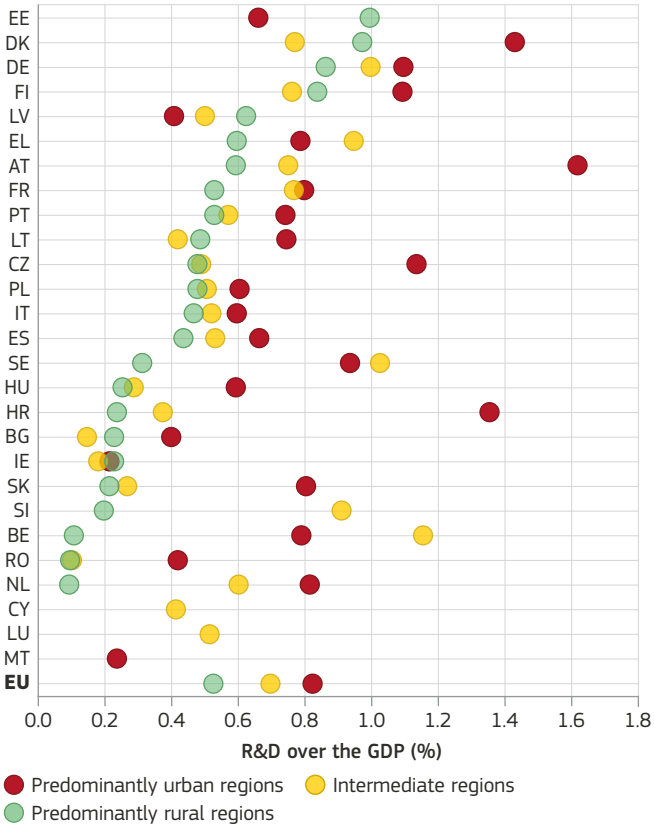
At the same time, territorial innovation capacity is more heterogeneous than aggregate figures suggest. Over 20% of rural regions perform above the overall EU R&D intensity average. This is particularly evident in rural regions of western Germany (e.g., Bautzen, Helmstedt), Upper Austria (e.g., Traunviertel, Innviertel) and south-western France (e.g., Lot, Tarn-et-Garonne), where R&D intensity exceeds 3% of GDP.

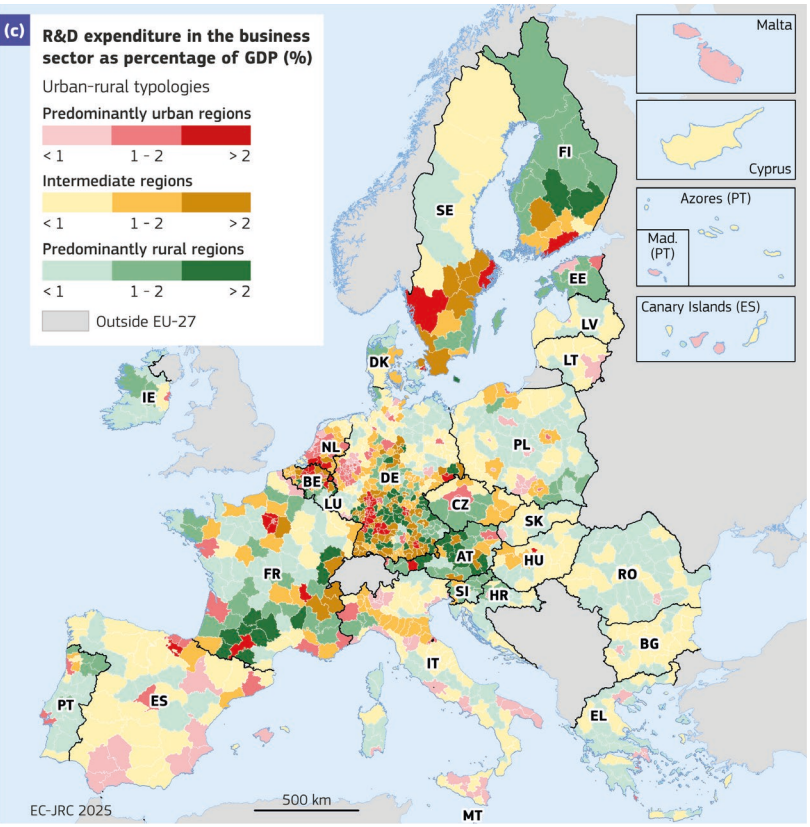
Figure 1 – Total **(a)**, public **(b)** and business **(c)** R&D expenditure as a percentage of GDP across EU regions by urban-rural typology



1 R&D expenditure at NUTS3 level is not available from Eurostat. Values refer to 2022 and are JRC estimates at NUTS3 level, classified by urban–rural typology, derived using a downscaling approach applied to NUTS2 data and informed by sectoral Gross Value Added (GVA), ensuring consistency with NUTS2 totals.







Source: JRC estimates based on Eurostat and ARDECO data.

Diverse mixes of public and business R&D underpin rural and urban innovation

Rural regions receive only around 12% of total EU R&D investments, despite hosting 21% of the EU population. They show lower levels of both public and business R&D investments, with **public R&D averaging 0.5% of GDP in rural regions** compared with 0.8% of GDP in urban regions and **business R&D averaging 1.1% in rural regions** as opposed to 1.6% in urban regions (Figures 1a and 1b). In a number of rural territories (e.g., in Germany, Austria, Belgium and Italy), public investment in research plays a key role in sustaining innovation potential and can partially compensate for more limited business R&D activity. Some of these cases reflect the role of public research institutions in enhancing innovation capacity in rural regions.

The landscape of innovative and creative outputs across EU regions

Urban regions dominate patenting, with pockets of rural excellence

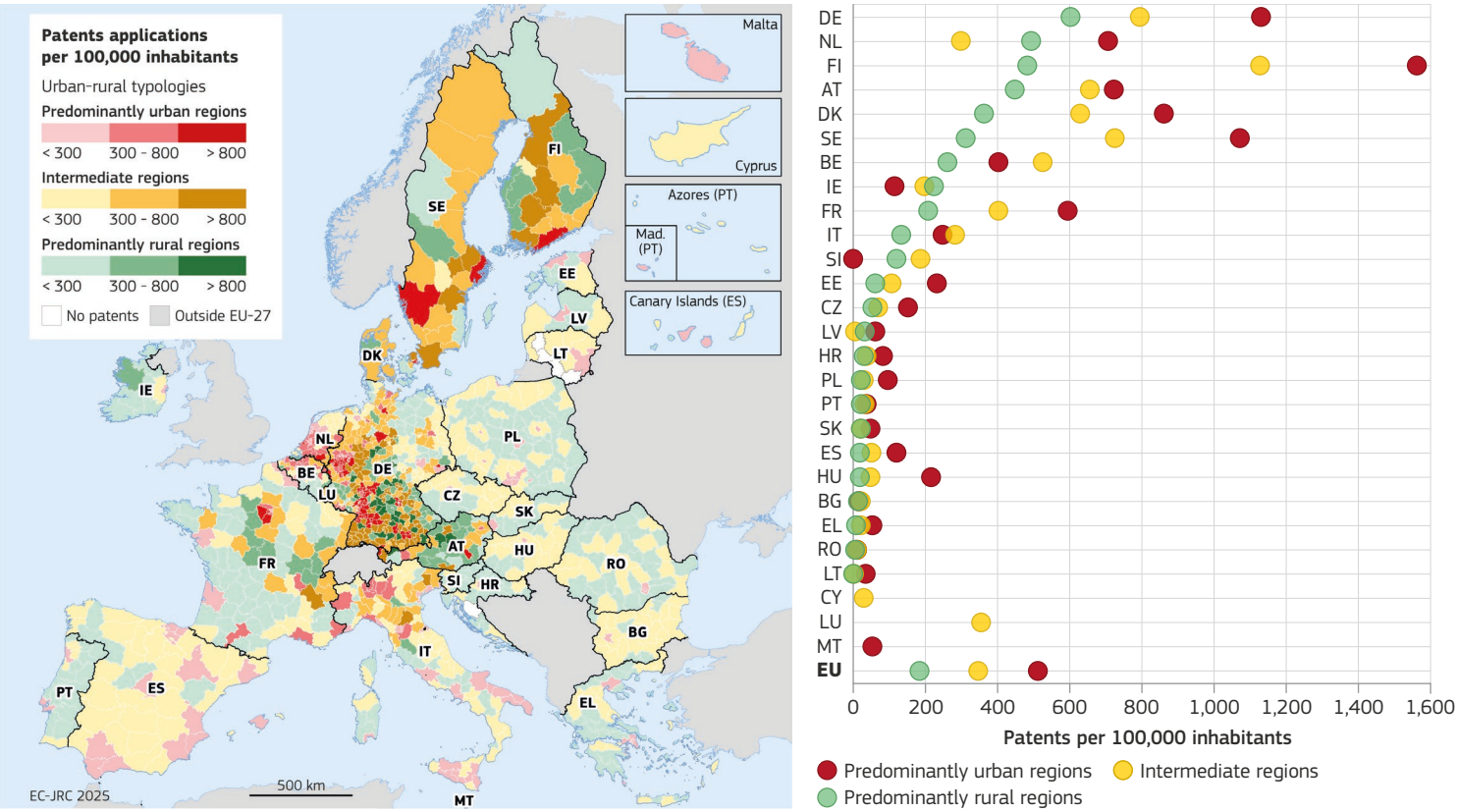
Patents, a core indicator of where technological innovation is developed, averaged around 380 applications per 100,000 inhabitants across the EU.[2] Germany is the undisputed EU leader in patent application, accounting for nearly 45% of all EU patent filings, far exceeding its share of EU population (around 19%).

At national level, population-adjusted figures reveal even sharper contrasts. Finland led with around 1,126 patents per 100,000 inhabitants, followed by Germany, Sweden, Denmark and the Netherlands, all exceeding 600 patents per 100,000 inhabitants, well above the EU average. By contrast, most Southern and Eastern countries remain below 100, highlighting persistent territorial divides in technological innovation capacity and the need for targeted support to strengthen it.

At the regional level, patent activity continues to be highly concentrated: more than 70% of EU filings originate from fewer than one-fifth of regions (around 220 NUTS3), which together account for about 36% of the EU population. Leading innovation hubs are located mainly in Central and Northern Europe, especially in large metropolitan and specialised industrial regions such as Erlangen (DE), Zuidoost-Noord-Brabant (NL), Munich (DE), Stockholm (SE), Helsinki (FI) or Paris (FR), each recording well above 1,200 patent filings per 100,000 inhabitants.

Territorial differences become even more evident when comparing rural, intermediate and urban regions. Urban regions show the highest patenting rates, averaging around 510 per 100,000 inhabitants, compared with about 350 in intermediate and roughly **180 patents per 100,000 inhabitants in rural regions** (Figure 2). Yet, several rural regions outperform the EU average. Hildburghausen (DE) stands out as Europe's top rural region with around 3,300 patent applications per 100,000 inhabitants. A number of rural regions in Finland, Germany, Austria, and the Netherlands perform above the EU urban average.

Figure 2 – Patent applications per 100,000 inhabitants across EU regions by urban–rural typology



Note: Patent counts refer to extended patent families filed by EU-based inventors and fractionally assigned to NUTS3 regions.
Source: JRC analysis based on EPO PATSTAT and Eurostat data.

2 Patent counts refer to extended patent families filed by inventors residing in EU Member States, as recorded in the 2023 Autumn edition of the EPO PATSTAT patent database. To avoid double counting, we assign families fractionally to NUTS3 regions based on the recorded residence of inventors. Following an extensive disambiguation and geolocation exercise, we identified a total of nearly 1.7 million patent families filed since 1990 attributed to EU-based inventors. This approach offers a key advantage over the OECD REGPAT, which is the other commonly used regionalised patent dataset. While REGPAT focuses on patents filed at the European Patent Office, PATSTAT draws on data from more than 100 national and regional patent authorities world-wide. This broader coverage, combined with our extensive disambiguation exercise, allows for a more complete capture of patenting activity, including that of smaller firms and organisations that tend to file at national offices, where procedures are generally more accessible and less costly.

These cases often reflect strong industrial specialisation and close linkages with nearby urban innovation hubs, showing that rural regions can also be sources of technological innovation when the right capabilities and urban-rural networks are in place.

Trademark activity is highest in cities, with standout rural regions featuring niche specialisations and cross-border business ecosystems

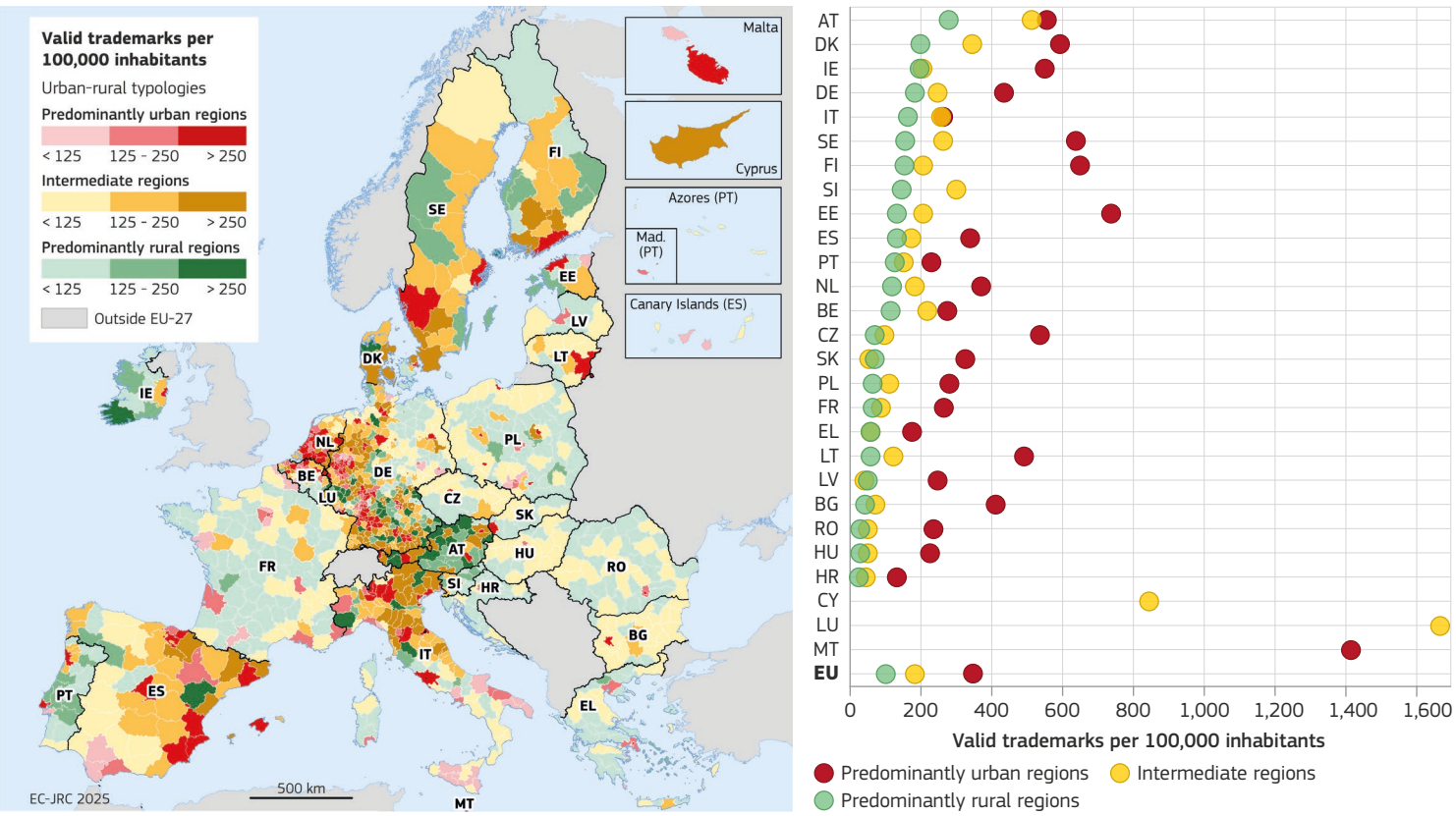
Trademark activity reflects where firms build customer recognition and often where they develop new products or enter new markets. As of June 2025, the EU holds a stock of nearly one million valid EU Trade Marks (EUTM), with Germany accounting for about 30% of the total stock (compared with 19% of EU population), followed by Italy (15% of EUTM vs 13% of EU population), Spain (13% vs 11%) and France (10% vs 15%).[3] On a per capita basis, the EU average stands at around 235 valid trademarks per 100,000 inhabitants. Smaller and more open economies perform particularly well, with Luxembourg (about 1,400 valid trademarks per 100,000 inhabitants), Cyprus (about 780), Estonia (about 460), and Austria (about 440), reflecting strong service sectors and international market ties.

At regional level, trademark activity is also highly concentrated, with most trademarks coming from a limited number of NUTS3 regions. Major urban and industrial centres lead the way, with Munich topping the list (1,623 trademarks per 100,000 inhabitants), followed by Paris, Düsseldorf, Amsterdam, Tallin, Salzburg, Milan, and Stockholm, each surpassing 800 trademarks per 100,000 inhabitants, supported by strong business ecosystems.

Only slightly over 20% of EU regions representing 27% of the EU population, mainly in western and southern Germany, Belgium, the Netherlands, Luxembourg, northern Italy, Austria, eastern Spain, and parts of the Nordic countries, exceed 300 trademarks per 100,000 inhabitants.

Trademark intensity shows a clear urban-driven pattern. Urban regions record the highest levels of trademarking, with around 346 trademarks per 100,000 inhabitants, compared with about 218 in intermediate regions and roughly **101 trademarks per 100,000 inhabitants in rural regions** (Figure 3). However, a number of rural regions perform above the EU average. Territories such as Cochem-Zell (DE), Åland (FI) Traunviertel (AT), Mantova (IT), South-West (IE), Vestjylland (DK) or Teruel (ES), show the presence of niche specialisations, or export-oriented activities and cross-border business ecosystems.

Figure 3 – Trademark stocks per 100,000 inhabitants across EU regions by urban-rural typology



Note: Trade mark figures refer to stocks of valid EU Trade Marks (EUTMs) held by EU-based applicants, as of June 2025. National filings are excluded.
Source: JRC analysis based on EUIPO and Eurostat data.

3 Trademark figures refer to the stock of valid EU Trade Marks (EUTMs) held by EU-based applicants, geolocated at NUTS3 level using applicant address information from EUIPO databases, as of June 2025. The indicator covers unitary EU rights only and excludes national trademark filings.

These cases demonstrate that market-driven innovation can emerge in rural regions when local assets are connected to broader economic networks.

While cities still lead, design activities reveal rural strengths in specialised and craft-based sectors

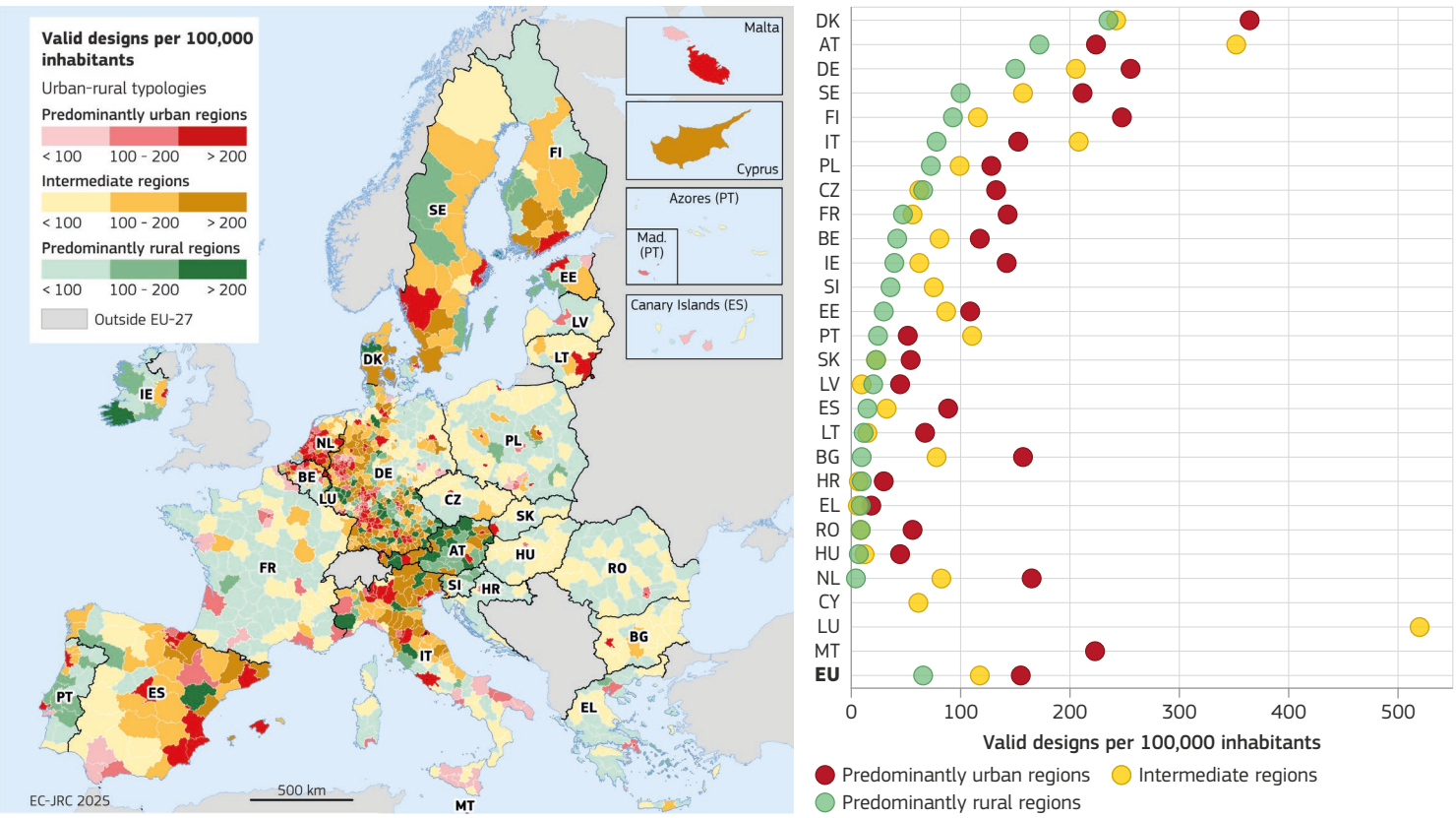
Industrial designs represent the creative side of industrial innovation. They focus on how products look, feel and function for users, strongly contributing to product differentiation and competitiveness. As of June 2025, a stock of around 551,000 valid Registered EU Designs (REUDs) was recorded in the EU, equivalent to an EU average of 123 valid designs per 100,000 inhabitants.[4] Luxembourg leads in per-capita terms (520 designs per 100,000 inhabitants) followed by Denmark, Austria, Malta and Germany, all exceeding 200 designs per 100,000 inhabitants.

Design activity is highly concentrated geographically, particularly in Central Europe, where about one-quarter of NUTS3 regions (home to approximately 43% of the EU population) hold three-quarters of the total stock of valid designs. This pattern differs from trademarks, which are more prominent in Southern and Baltic Europe, pointing to distinct specialisations across innovation outputs. Leading hubs include Rosenheim Kreisfreie (DE), Tiroler Unterland (AT), Erlangen-Höchstadt (DE), Stuttgart (DE), Rottweil (DE), Paris (FR) or Zuidoost-Noord (NL), all exceeding 800 valid designs per 100,000 inhabitants. These regions combine competitive industrial bases with vibrant creative sectors. Nordic regions, notably in Sweden, Finland and Denmark, also exhibit high design per-capita intensity.

Design intensity reveals an urban–rural gradient. On average, **urban regions record around 156 designs per 100,000 inhabitants,** compared with 118 in intermediate and **56 in rural regions** (Figure 4).

Leading urban regions include Erlangen-Höchstadt and Stuttgart (DE), Paris (FR), Rimini and Milan (IT), Zuidoost-Noord (NL), and Rheintal-Bodenseegebiet (AT). By contrast, around four in five (82%) EU rural regions perform well below the EU average. Yet, several rural regions outperform, including Steyr-Kirchdorf (AT), Pirmasens, Karlsruhe-Stadt, and Main-Tauber-Kreis (DE), Vestjylland (DK), Tarnowski (PL) and Vercelli (IT) – all exceeding 400 valid designs per 100,000 inhabitants. These territories often combine niche manufacturing and craft-based industries with strong design traditions. For instance, Vercelli, known for its large rice-growing areas, also specialises in textile and fashion-related design, illustrating how also rural regions with significant agricultural activity can diversify into design-intensive manufacturing.

Figure 4 – Industrial design stocks per 100,000 inhabitants across EU regions by urban-rural typology



Note: Design figures refer to stocks of valid Registered EU Designs (REUDs) held by EU-based applicants, as of June 2025. National filings are excluded.
Source: JRC analysis based on EUIPO and Eurostat data.

4 Industrial design figures refer to the stock of valid Registered EU Designs (REUDs) held by EU-based applicants, geolocated at NUTS3 level using applicant address information from EUIPO databases, as of June 2025. The indicator covers unitary EU rights only and excludes national design filings.

Clusters of intermediate regions across northern Italy, western and southern Germany, Austria, and Denmark also perform well, likely benefiting from proximity to urban hubs and specialised industrial networks. Notably, Rosenheim Kreisfreie (DE) and Tiroler Unterland (AT) show the highest per capita design activity in Europe (3,252 and 2,051 valid designs per 100,000 inhabitants, respectively).

A design-to-trademark ratio provides further insights into regional specialisation. Most EU regions (81%) are more branding-oriented (ratio<0.8), especially in Southern and Eastern Europe. A smaller group (9.5%) of design-oriented regions (ratio>1.2) –including Rosenheim Kreisfreie, Tiroler Unterland, Erlangen-Höchstadt, Treviso, and Jönköpings län – demonstrates strong industrial clusters and design-led innovation capacity. Balanced ecosystems (0.8 < ratio < 1.2) are mainly found in Central European and Polish regions.

Turning innovation potential into opportunity in all regions

Innovation represents a key engine of both competitiveness and territorial cohesion in the EU. The metrics presented in this brief – for the first time constructed and estimated at a sufficiently granular territorial level to capture differences between urban and rural regions – reveal that innovation capacity continues to be unevenly distributed across European regions. Urban areas concentrate most R&D efforts and innovative outputs protected by intellectual property rights. However, the evidence also clearly reveals strong heterogeneity both across and within countries. A significant number of rural and intermediate regions demonstrate strong technological, branding and design performances. These cases show the importance of urban-rural linkages, public–private cooperation, and tailored support in helping rural territories activate their potential and overcome structural disadvantages.

Going forward, innovation policies should better reflect territorial diversity, supporting excellence where it emerges, while creating enabling conditions where gaps remain. Key policy directions (e.g., European Commission, 2022a; 2021b; 2023; EUIPO & EPO, 2025) include:

- Leveraging territorial data and intelligence to guide evidence-based policymaking, track regional innovation dynamics, and strengthen support where it is most needed.
- Strengthening place-based innovation ecosystems in rural and lagging regions, by investing in skills, infrastructure, digital connectivity, and ecosystem coordination.
- Boosting the uptake of intellectual property rights, especially among micro-firms, SMEs, and startups, through IP awareness initiatives beyond urban cores, targeted advisory services and simplified access to IP tools.
- Leveraging public R&D investment to unlock complementarities with business efforts, particularly where universities, research centres or demonstration facilities can anchor local capacity.

Looking ahead, further work should focus on improving data granularity and timeliness to better capture territorial innovation dynamics. The JRC is actively advancing in this direction, designing metrics that capture not only innovation creation but also adoption in businesses – which is particularly relevant to boosting competitiveness in rural regions – and broadening the focus beyond technological innovation to include process and organisational innovation.

Finer, broader and more consistent innovation metrics are needed to help policymakers identify emerging opportunities and complementarities, tailor support where it is most needed and ensure that innovation becomes not only a driver of competitiveness, but also a foundation for cohesion and shared prosperity across all European regions.

Box 1: Economic relevance of trade-marks and designs in the EU

Intellectual property rights (IPR)-intensive industries are a cornerstone of the EU economy. Between 2021 and 2023, they generated 48% of EU GDP and provided nearly 31% of all jobs. Within this group, trademark-intensive industries alone contributed 39% of GDP and 22% of employment. Design-intensive industries contributed another 16% of GDP and 13% of jobs, while patent-intensive industries accounted for 18% of GDP and 12% of employment.

At the firm level, evidence from 2013–2022 shows that companies with IPRs generate 24% more revenue per employee and pay 22% higher wages than those without. After adjusting for industry, size and location, the revenue advantage rises to 41%, with even stronger effects for SMEs: those registering IPRs report 44% higher revenue per employee. Yet, only 10% of SMEs hold IPRs, compared with about half of large companies. The latest Economic Performance of IPR Indicators confirms a strong post-COVID rebound in trademark and design filings, but also highlights persistent regional disparities across the Union.

Sources: EUIPO & EPO (2025, 2026); EUIPO (2023)

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